THE SIGNALRUPTURE FROZEN MEASUREMENT PROTOCOL

Pre-Registered Operational Definitions for Empirical Testing of the SR Structural Grammar

SignalRupture (SR)

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Abstract

This manuscript freezes the measurement architecture for the first empirical program of SignalRupture (SR) as a developing structural metatheory. Its purpose is to prevent post-hoc movement of variables, thresholds, domains, and outcomes after results become known. The protocol operationalizes the SR structural grammar through six positions: underlying condition (X), representation (R), effective requirement (Q), durable capacity (D), sustainable flexibility (C_S), extraordinary compensation (C_E), and observable function (F). It specifies measurement rules for Canadian healthcare, North American electricity, and Canadian households; defines the principal longitudinal hypotheses; separates ordinary resilience from extraordinary compensation; establishes negative controls and restoration cases; and freezes statistical comparison rules. The primary empirical question is not whether stressed systems use buffers. Established fields already show that they do. The SR question is whether the state and trajectory of compensation provide incremental, prospectively useful information about hidden requirement–capacity mismatch and subsequent functional deterioration or restoration across heterogeneous domains. No result may be called supportive merely because a plausible mapping can be constructed after the outcome is observed.

Keywords: SignalRupture; structural grammar; preregistration; measurement protocol; compensated continuity; compensatory inversion; healthcare; electricity; households; falsification.

1. Purpose and Freeze Rule

The protocol is frozen in August 2026 before the principal prospective observation windows. Future revisions may create new versions, but they may not overwrite this registry. A variable may be corrected for documented data-quality reasons, but its grammatical role may not be changed merely because the original specification fails.

$$\text{Frozen mapping} + \text{future observation} = \text{prospective test}$$

Every empirical paper must report deviations from this protocol. A deviation can be scientifically justified, but the original test must remain scoreable.

2. Canonical Grammar

Symbol	Frozen role	General measurement rule
X	Underlying condition	Domain state independently measured from the focal representation.
R	Representation	Official metric, forecast, classification, dashboard, or institutional representation of X.
Q	Effective requirement	Quantity/configuration required to maintain a predeclared function or target.
D	Durable capacity	Sustainable ordinary capacity available without extraordinary/depleting support.
C_S	Sustainable flexibility	Designed, renewable buffering expected under normal operation.
C_E	Extraordinary compensation	Temporary, unusually costly, difficult-to-replenish, emergency, or depleting bridge capacity.
F	Observable function	Predeclared output, access, continuity, reliability, or social-reproduction outcome.

$$G_{C,t} = Q_t - D_t$$

$$\text{Compensated Continuity: } G_{C,t} > 0 \wedge C_{E,t} \uparrow \wedge F_t \approx \text{stable}$$

$$\text{Compensatory Inversion: } \Delta C\_E,t \leq 0 \wedge G\_C,t > 0 \wedge \Delta F\_t < 0$$

3. Universal Classification Rules

• A variable counts as Q only if a target or service requirement is declared before analysis.

• A variable counts as D only if it can be sustained under ordinary operation for the relevant horizon.

• A variable counts as C_E only if evidence shows that it bridges a Q–D mismatch and is exceptional, costly, constrained, temporary, or difficult to replenish.

• High utilization alone does not classify a resource as extraordinary compensation.

• F must be measured independently of C_E wherever possible to avoid circularity.

• F is multidimensional when volume and access can diverge; no single headline measure may erase a conflicting functional outcome.

• Falling C_E is not inversion when D rises, Q falls, or F improves.

• No universal scalar combining incompatible domain units will be constructed in the first validation phase.

4. Primary Domain I — Canadian Healthcare

Healthcare is the primary empirical domain because extraordinary human compensation is directly observable and because provincial and regional heterogeneity creates comparison opportunities.

Grammar role	**Frozen healthcare measure**	**Secondary/sensitivity measures**
Q	Service/workforce requirement implied by realized demand and predeclared access/service targets	Population aging; ED visits; admissions; surgical demand; occupancy; referral volume.
D	Regular sustainable staffed capacity	Regular paid hours; staffed beds; permanent FTEs; active workforce; effective new entrants minus exits.

C_S	Normal scheduled flexibility	Historically normal overtime band; ordinary casual/float coverage where demonstrably renewable.
C_E	Overtime above normal band + purchased/agency staffing + emergency redeployment where measured	Cost per purchased hour; repeated extra shifts; temporary closures avoided through extraordinary staffing.
F_volume	Delivered service volume	Admissions; procedures; ED visits completed.
F_access	Access/latency	ED waits; surgery waits; benchmark attainment; referral backlog; time to primary/specialist care.
F_discontinuity	Observable service loss	Closures, diversions, cancelled services, persistent rationing by delay.
HB depletion indicators	Human-buffer condition	Absence, sick leave, turnover, vacancy duration, retention, disability leave, retirement/exit.

The primary healthcare unit should be the smallest stable geographic/service unit for which Q, D, C_E and F can be observed repeatedly. Province-year is acceptable for initial replication but insufficient for the strongest causal test. Region, hospital system, occupation, or service-line panels are preferred.

5. Healthcare Frozen Outcomes

Primary outcomes are frozen in this order:

Priority	**Outcome**	**Direction interpreted as deterioration**
1	Access latency	Longer waits / lower benchmark attainment
2	Persistent service discontinuity	More closures, diversions, cancellations
3	Backlog	Faster accumulation / slower clearance

4	Workforce durability	Higher vacancy duration, absence, turnover
5	Service volume conditional on demand	Lower delivered service relative to realized demand

Service volume alone may not classify the system as restored if access latency, backlog, or discontinuity worsens. This rule is frozen to prevent headline output from dominating the substrate test.

6. Healthcare Compensation Baseline

The normal overtime band must be estimated from a pre-stress reference period where available, preferably 2015–2019, stratified by occupation and setting. If comparable pre-2019 data are unavailable, the earliest stable multi-year period must be used and documented. Purchased staffing is not automatically extraordinary; its historical baseline and renewability must also be estimated.

$$C\_E = \max(0, \text{Overtime} - \text{Normal Overtime Band}) + \text{Extraordinary Purchased Staffing} + \text{documented emergency labour}$$

Alternative definitions must be reported as sensitivity analyses, not substituted silently for the frozen primary definition.

7. Primary Healthcare Hypotheses

ID	Frozen hypothesis	Falsifying pattern
H1	Positive G_C is associated with greater subsequent C_E dependence.	No association or opposite association after demand controls.
H2	C_E attenuates contemporaneous deterioration in F under positive G_C.	C_E adds no explanatory information or is associated only with simultaneous deterioration.
H3	C_E trajectory improves prediction of F at t+h beyond F, Q, D and sector controls.	No out-of-sample improvement.

H4	CI predicts subsequent deterioration more strongly than overtime level alone.	CI performs no better than simple overtime/vacancy/wait baselines.
H5	Restoration—D rising relative to Q—reduces C_E dependence while F stabilizes/improves.	C_E remains necessary despite measured closure of G_C.

8. Primary Domain II — North American Electricity

Grammar role	**Frozen electricity measure**
Q	Realized/forecast peak demand plus reliability requirement under the relevant planning criterion.
D	Dependable/accredited resource capacity deliverable within the region, including firm transmission capability where planning rules count it.
C_S	Planned reserve, normal demand response, storage cycling, routine imports and other renewable flexibility already embedded in adequacy planning.
C_E	Emergency imports, emergency demand response, conservation appeals, exceptional reserve depletion, temporary waivers, load shedding avoidance measures outside normal planning assumptions.
F	Reliable delivered electricity: adequacy events, scarcity conditions, involuntary curtailment, EEA events, unserved energy where available.
Restoration	Commissioned dependable generation, transmission, storage/flexibility or demand reduction that durably reduces G_C.

Reserve margin itself is not extraordinary compensation. This is a frozen negative classification derived from the initial electricity probe. SR must not pathologize normal reliability engineering.

9. Electricity Hypotheses

ID	Frozen hypothesis	Falsifier
E1	Regions with shrinking G_C margin rely more heavily on C_E under comparable stress.	No relationship after weather/outage controls.
E2	C_E preserves F during temporary Q>D episodes.	No measurable buffering effect.
E3	Persistent reliance on C_E without durable additions predicts greater future scarcity/discontinuity risk.	Established adequacy measures fully explain outcomes and C_E adds no value.
E4	Durable additions that close G_C reduce subsequent C_E reliance and risk.	Risk does not decline after verified capacity restoration.

10. Primary Domain III — Canadian Households

Households are explicitly classified as the difficult domain. The first phase will not treat aggregate debt as compensation. Compensation must be linked to an identified essential-resource gap at the household or pre-specified subgroup level.

Grammar role	Frozen household measure
Q	Essential household requirement: shelter, food, utilities, transportation required for work, essential care, and minimum debt obligations under a declared budget standard.
D	After-tax disposable resources available from ordinary income plus stable recurring transfers, net of unavoidable taxes/withholdings.
C_S	Liquid savings intended for smoothing, ordinary family risk-sharing, and other demonstrably renewable buffers.
C_E	Savings depletion beyond renewable level, borrowing used for essentials, arrears deferral, additional paid work undertaken to close essential gap, exceptional family transfers, delayed essential care where measured.

F	Maintenance of housing, food security, utilities, essential care, solvency and minimum social reproduction.
Discontinuity	Food insecurity, eviction/homelessness, utility disconnection, insolvency, persistent arrears, unmet essential care, involuntary crowding.

Mortgage balances, total consumer credit, and debt-to-income ratios cannot independently establish C_E. They are contextual variables unless the data identify the purpose and sequence of borrowing.

11. Household Hypotheses

ID	**Frozen hypothesis**	**Falsifier**
HH1	Households with Q>D are more likely to deploy identified C_E than matched households without the gap.	No difference after income/composition controls.
HH2	C_E initially delays functional discontinuity among Q>D households.	No buffering period or worse outcomes immediately without plausible selection explanation.
HH3	Depletion/constraint of C_E predicts later deprivation or insolvency under persistent Q>D.	No prospective relationship.
HH4	When D rises or Q falls enough to close the gap, C_E use and deprivation should decline.	No recovery relationship.

12. Representation-Rupture Measurement

Representation rupture is tested separately from capacity grammar. Each study must name a representation R and an external condition X that R is expected to track. The analyst must state the expected correspondence before testing.

$$G_{R,t} = d(X_t, R_t)$$

Test	Frozen requirement
Calibration	Does R retain its expected empirical relationship with X?
Divergence	Does the R–X relationship weaken or reverse during the observation period?
Decision relevance	Does adding X/substrate indicators improve prediction or decisions beyond R?
Falsifier	R remains calibrated and alternative substrate measures add no meaningful information.

13. Statistical Model Freeze

$$M0: F\_(i,t+h) = \alpha + \beta_1 F\_(i,t) + \beta_2 Q\_(i,t) + \beta_3 D\_(i,t) + \gamma Z\_(i,t) + u\_i + \tau\_t + \varepsilon\_(i,t)$$

$$M1: M0 + \beta_4 C\_E,(i,t) + \beta_5 \Delta C\_E,(i,t) + \beta_6 [G\_C,(i,t) \times \Delta C\_E,(i,t)]$$

M0 is the sector baseline. M1 is the SR augmentation. Z contains predeclared domain controls; u_i and τ_t represent unit and time effects where appropriate. The exact estimator may change with outcome type, but the comparison between a credible sector baseline and an SR-augmented model is mandatory.

For binary discontinuity outcomes, logistic, complementary log-log, or survival/event-history models may be used. For continuous latency/backlog outcomes, panel or time-series models are appropriate. Where latent HB is modeled, confirmatory factor/state-space specifications must be validated separately before predictive use.

14. Prediction and Validation Rules

- Temporal split or rolling-origin validation is mandatory where sample size permits.
- The primary claim concerns out-of-sample performance, not in-sample R^2 alone.
- Report calibration and discrimination for event predictions.
- Report MAE/RMSE or appropriate scoring rules for continuous outcomes.

- Compare against simple baselines: lagged F, Q–D gap alone, overtime/compensation level alone, and the strongest established sector model available.
- Confidence intervals and effect sizes must be reported; p-values alone are insufficient.
- Missing data handling must be specified before outcome inspection where possible.
- Multiple outcome testing must be transparently reported; primary outcomes remain prioritized as frozen above.

15. Compensatory Inversion Event Rule

A CI event is not declared from a single period unless the domain's natural frequency makes that appropriate. The primary rule requires persistence across at least two consecutive measurement intervals, except for annual data where one interval may be provisionally flagged and must be confirmed at the next observation.

$$\mathrm{CI} = \text{persistent } [\Delta C\_E \leq 0] + \text{persistent } [G\_C > 0] + \text{worsening predeclared } F$$

If C_E falls because ordinary capacity rises, demand/requirement falls, substitution becomes durable, or outcomes improve, the observation is scored Restoration/Normalization, not CI.

16. Negative Controls

Control case	Why required
High C_S, stable D≥Q, stable F	Shows that normal buffering is not misclassified as rupture.
High C_E followed by rapid D restoration and improving F	Tests the recovery branch.
F deterioration without positive G_C	Tests whether SR incorrectly explains every failure through capacity deficit.
Positive G_C with no C_E and no F deterioration	Tests whether the declared requirement or durable-capacity measure is misspecified.
Stable representation R with strong X calibration	Negative control for representation rupture.

17. Confounding and Alternative Explanations

Every domain analysis must explicitly model plausible alternatives. Healthcare examples include demand shocks, wage policy, pandemic effects, demographic change, scope-of-practice changes, technology, funding, and regional labor markets. Electricity examples include weather, forced outages, fuel availability, transmission constraints, interconnection, market design, and forecast error. Household examples include household composition, regional housing market, interest rates, unemployment, inflation, immigration status only where ethically and statistically appropriate, and life-cycle borrowing.

SR is not supported when an omitted conventional mechanism explains the apparent grammar more parsimoniously.

18. Cross-Domain Confirmation Standard

No single domain validates the metatheory. The first empirical threshold is frozen as follows:

Requirement	Minimum
Domains	Three, including at least one predominantly engineered/non-human system.
Mapping	Pre-specified Q, D, C_E and F in each.
Time structure	Longitudinal/prospective rather than cross-sectional only.
Prediction	At least two domains show incremental out-of-sample value from compensation state.
Restoration	At least one clear case correctly follows the restoration branch.
Failure opportunity	At least one domain or hypothesis is allowed to fail without redefining the grammar.
Replication	At least one independent or separately assembled dataset reproduces a principal result.

Meeting this threshold constitutes empirical support for the structural grammar, not proof of universality.

19. Frozen Interpretation Categories

Score	Meaning
Supported	Pre-specified direction/mechanism observed and SR augmentation adds meaningful value.
Partially Supported	Direction appears but mechanism, measurement, timing, or incremental value is incomplete.
Indeterminate	Data or observation window insufficient.
Not Supported	Predicted relationship absent without a compelling measurement failure.
Falsified	Pre-specified countercondition occurs or repeated tests contradict the relation.

20. Data Provenance Standard

- Prefer official administrative or statistical sources for primary measures.
- Record release date, revision status, geography, unit, frequency, denominator, and methodological breaks.
- Do not splice incompatible series without documenting harmonization.
- Archive the exact extraction used for each analysis where licensing permits.
- Freeze transformations and exclusions in code or a public analysis plan before final outcome testing.
- Retain null results and superseded estimates.

21. First Formal Study — Frozen Design

The first formal study is Canadian healthcare. Baseline construction should use the longest comparable pre-2026 series available, with 2026 designated as the canonical prospective baseline for the post-gauntlet SR program. The principal prospective saturation window remains 2027–2029. Earlier historical data may be used for model development and retrospective validation, but must not be presented as if they were prospective predictions.

The primary comparison will test whether compensation state and trajectory improve prediction of access latency, discontinuity, backlog, and workforce durability beyond lagged outcomes, demand, ordinary staffing capacity, and standard sector controls. The strongest version uses regional/hospital/occupation panels; province-level analysis is an initial replication layer rather than the endpoint.

22. Second Formal Study — Frozen Design

The second study is electricity. It will compare regions with different realized load growth, dependable resource additions, and restoration lead times. The study must preserve NERC/ISO reliability definitions and must not substitute SR terminology for technical adequacy measures. SR succeeds only if its compensation/restoration representation adds decision or predictive value beyond those established measures.

23. Third Formal Study — Frozen Design

The third study is households and should not proceed to a strong CI claim until longitudinal microdata can identify the same household's essential-resource gap, coping response, and later functional outcome. Aggregate national debt statistics may contextualize the study but cannot validate the compensatory sequence.

24. Frozen Primary Claims

ID	Claim
P1	Observable continuity does not by itself establish durable structural sufficiency.

P2	Under positive Q–D mismatch, identifiable compensation can preserve function temporarily.
P3	The state and trajectory of extraordinary compensation may improve prospective inference about later function.
P4	Compensation decline is a warning only when the structural gap persists and function worsens.
P5	Durable restoration should reduce both the gap and extraordinary compensation dependence while maintaining/improving function.
P6	The same logical grammar should translate across heterogeneous domains without post-hoc role changes.

25. Claims Explicitly Not Frozen as Established

- Compensatory Inversion is not yet an empirically established theorem.
- Human Buffer is not yet a validated universal latent variable.
- There is no calibrated universal collapse probability.
- There is no universal numerical threshold for Q–D deficit.
- There is no universal scalar combining healthcare, households, electricity, governance, or digital systems.
- Cross-domain support does not establish causality without domain-specific identification.
- SR does not replace resilience engineering, reliability analysis, public administration, economics, epidemiology, or other domain sciences.

26. Prospective Registry

ID	Window	Test	Outcome

FMP-1	2026–2029	Healthcare G_C → C_E dependence	Association and predictive increment
FMP-2	2027–2029	Healthcare CI emergence	Access/workforce deterioration after persistent CI
FMP-3	2026–2032	Electricity requirement/capacity race	C_E reliance and adequacy outcomes
FMP-4	2026–2032	Electricity restoration branch	Risk declines after verified durable additions
FMP-5	2027–2031	Household compensation sequence	Gap → identified coping → later deprivation
FMP-6	2026 onward	Representation rupture	R–X divergence adds decision-relevant information
FMP-7	After ≥3 completed domains	Cross-domain grammar	Incremental predictive value and replication

27. Canonical Empirical Standard

SR will not call the Structural Grammar empirically validated merely because examples can be mapped onto it. Empirical support requires frozen operationalization, longitudinal observation, explicit comparison with established models, negative cases, restoration cases, and out-of-sample prediction. Cross-domain recurrence is evidence only when the grammatical roles remain stable before outcomes are known.

Conceptual fit < retrospective association < prospective prediction < replicated incremental value

28. Conclusion

The purpose of this protocol is to make SignalRupture vulnerable to evidence. The Structural Grammar becomes scientific only when its variables can be measured without circularity and when its predictions can lose. The first validation phase therefore freezes the meanings of requirement, durable capacity, sustainable flexibility, extraordinary compensation, and observable function across three deliberately different domains. Healthcare offers direct human compensation measures; electricity provides an engineered-system challenge; households provide a difficult social test in which compensation cannot be inferred casually from debt.

The central empirical burden is narrow. SR must show that compensation state reveals something about structural sufficiency and future function that established sector measures do not already reveal. If it does so prospectively and repeatedly across domains, the grammar gains empirical metatheoretical support. If it does not, the grammar must narrow, converge with established theory, or be rejected. This protocol freezes that burden in August 2026.

Appendix A. Minimal Data Schema

Field	Required
unit_id	Stable geographic/system/household identifier
time	Quarter/year or domain-appropriate interval
Q	Effective requirement
D	Durable capacity
C_S	Sustainable flexibility
C_E	Extraordinary compensation
F_primary	Primary functional outcome
F_secondary	Secondary functional outcomes
X	Underlying condition where representation test used
R	Representation where representation test used
controls	Predeclared domain covariates

source/version	Provenance and revision metadata

Appendix B. Analysis Output Template

• Descriptive trajectories of Q, D, G_C, C_S, C_E, and F.

• Pre-specified classification of compensated continuity, restoration, CI, or neither.

• Baseline model performance.

• SR-augmented model performance.

• Out-of-sample performance difference with uncertainty.

• Alternative explanations and robustness checks.

• Negative-control results.

• Forecast score: Supported / Partially Supported / Indeterminate / Not Supported / Falsified.

• Deviation log from this frozen protocol.

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